Noushin Behboudi

(+1) 240-446-9454 | noushin7@vt.edu | GitHub

Work Experience

Research Technical Staff

George Mason University, Fairfax, VA — *Apr 2020 - Jul2025*

- Developed and migrated an outdated, server-side-rendered web application (the NSF-funded Paleobiology Database: www.paleobiodb.org) to a full-stack architecture by enhancing the interface and implementing client-side rendering with JavaScript, Bootstrap, and jQuery, resulting in improved performance, maintainability, and user experience
- Developed a Python application using SQLAlchemy with SQLite to update bibliographic data for 89,500 records in the Paleobiology Database (the leading scientific research tool), ensuring accuracy and consistency
- Developed a Python script to automate scanning NetCDF files, extracting metadata, and generating Intake YAML catalogs and HTML pages for streamlined data management, discovery, and visualization
- Developed and maintained an open-source web-based data cataloging system for Earth system science datasets, enabling efficient management and organization of large, shared climate datasets across multiple users without centralized coordination. Designed to be adaptable for any dataset collection.
- Developed and maintained scripts to process and analyze climate datasets, ensuring accurate data preparation and seamless integration into research and modeling pipelines
- Built web applications using Flask, Python and auto-deployed on Heroku via GitHub

Data Science Fellow

Metrostar Systems, Reston, VA — *Oct 2019 - Apr 2020*

- Developed machine learning Proof-of-Concepts to help drive targeted business questions, applying existing methods or developing new methods
- Implemented a *serverless* architecture using API Gateway, Lambda, and RDS and deployed AWS Lambda code from Amazon S3 buckets
- Created a Lambda Deployment function, and configured it to interact with DynamoDB
- Used Postman to test RESTful APIs
- Implemented Swagger/Swagger UI to streamline endpoint documentation
- Developed web scrapers using Selenium and BeautifulSoup to automate data collection from five websites, feeding real-time data into a machine learning tool for analysis and insights

Python Developer

Virginia Tech — *Jun. 2018 - Aug. 2018*

- Converted the implementation of an impedance extraction algorithm from Matlab to Python using Numpy library

Software Developer

Tosan Intelligent Data Miners — *Jan. 2014 - Mar. 2016*

- Developed highly efficient algorithms in C++ specializing in the development of a transaction processing infrastructure to provide fraud detection services

Certificates

Supervised Machine Learning: Regression and Classification – deeplearning.ai, November 2024

Programming & Software Skills

Programming:	Python, C/C++, JavaScript, MATLAB, Simulink, SQL, Bash shell scripting
Orchestration Tools	Apache Airflow (Familiar)
Artificial Intelligence & Machine Learning	Self-Supervised Learning Networks, Vision Transformer (ViT), Swin Transformer, Machine Learning Algorithms, PyTorch, MONAI. Implemented SimMIM (Self-supervised Masked Image Modeling) using the MONAI for medical imaging experiments Familiarity with Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG)
Cloud:	AWS EC2, RDS, DynamoDB, S3, Lambda, API Gateway
Libraries	NumPy, Matplotlib, Scikit-learn, Pandas, Beautiful Soup, , xarray, NLTK, TensorFlow, Boto3, SQLAlchemy
Technology Experience:	Unix/Linux development tools (GDB, GCC, Make, GNU Autotools (Automake, Autoconf, and Libtools))
Databases:	MySQL, MariaDB, SQLite
Containers and Orchestration:	Docker, Kubernetes (Familiar)
Frameworks:	Flask
Version control systems:	Git/GitHub/SourceTree
Productivity Applications:	Vim, Microsoft Office Suite
Web Technologies:	HTML/HTML5, CSS3, jQuery, AJAX, Bootstrap, JSON, XML/YAML
Web Service:	RESTful, Postman
Technologies/Tools:	TEX (LATEX), Swagger, Sphinx, Selenium, Jekyll, Markdown, GitHub Pages

Education

Master of Science, Trine University
INFORMATION STUDIES *2025--Present*

Master of Engineering, Virginia Tech
COMPUTER ENGINEERING *2017-2019*

- GPA: 3.75

Master of Science, University of Tehran
COMPUTER ENGINEERING *2009-2012*

- Thesis Title: Power-Aware Application Specific Instruction-Set Design
- GPA: 3.63

Bachelor of Science, Iran University of Science and Technology
COMPUTER ENGINEERING *2004-2009*

- *Honor student*